Envisage: Depth-Conditioned Diffusion Inpainting for Facial Surgery Outcome Prediction

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Abstract. Predicting the visual outcome of facial surgery from a single pre-operative photograph is important for patient consent and surgical planning, yet existing approaches either apply geometric morphing that lacks photorealism or generate the entire face from a conditioning signal, altering patient identity in regions the surgeon does not touch. Here, we mask only the surgical region and modify a monocular depth map to reflect the target tissue displacement, then use a pretrained depth-conditioned diffusion model to regenerate the masked area. We introduce decomposed identity evaluation, in which ArcFace cosine similarity is measured separately on the surgical region, non-surgical region, and full face. On the HDA Plastic Surgery benchmark (637 pairs, 104 test, 3 procedures), Envisage achieves 0.958 ArcFace for blepharoplasty (compared to 0.670 for LandmarkDiff, a full-face diffusion baseline that we show relied on compositing for 96% of its identity score), 0.811 for rhytidectomy (compared to 0.360), and 0.725 for rhinoplasty (+19%), with non-surgical region ArcFace of 0.922 to 0.978. These results are obtained with zero task-specific training; the pretrained depth ControlNet suffices to produce anatomically coherent predictions from modified depth maps alone.

Keywords: Facial surgery prediction · Diffusion inpainting · Depth conditioning · Identity preservation · Fairness

1 Introduction

Facial plastic surgery accounts for over 15 million procedures per year globally, with rhinoplasty and blepharoplasty among the most requested [1]. Patient satisfaction is strongly associated with pre-operative expectation management; studies report revision rates of 5 to 15% for rhinoplasty, driven in part by discrepancies between expected and actual outcomes [2,3]. Current prediction tools range from surgeon-drawn sketches to proprietary 3D simulation systems (Crisalix, Vectra 3D) that require dedicated hardware costing $30,000 to 50,000. No widely available system produces photorealistic predictions from a single 2D photograph while guaranteeing that the patient's identity is preserved in non-surgical regions.

Several approaches to automated surgical prediction have been explored. Geometric morphing methods based on thin-plate spline (TPS) warps [12] can approximate structural changes but introduce boundary artifacts and cannot synthesize new skin texture. Three-dimensional approaches (ACMT-Net [16], Ma

et al. [15]) operate on CT scans or 3D meshes, limiting their applicability to the 2D photograph setting that is most accessible in clinical practice. Full-face diffusion models regenerate the entire face from a structural conditioning signal. Our earlier system, LandmarkDiff, used SD 1.5 conditioned on landmark wireframes with post-hoc compositing. While it established the problem formulation and evaluation framework for diffusion-based surgical prediction, it achieved ArcFace 0.607 for rhinoplasty with compositing, and decomposition revealed that nearly all of that score was attributable to composited pixels rather than the diffusion model itself (0.023 without compositing). Envisage addresses every root cause identified in that work. DragDiffusion [13] allows point-based image editing but provides no procedure-specific anatomical structure.

These observations motivated a fundamental redesign. Rather than generating the full face and compositing afterward, we formulate surgical prediction as *inpainting*: mask the surgical region, modify the depth map to reflect the target tissue change, and let a pretrained depth-conditioned diffusion model regenerate only the masked area. The key insight is that depth modification (e.g., flattening the nasal bridge depth to simulate dorsal hump reduction) provides a physically meaningful conditioning signal that maps to actual tissue displacement during surgery. We use FLUX.1-dev [8] with a pretrained depth ControlNet [6]; no task-specific training is required for the core pipeline.

Our contributions are as follows.

1. An inpainting-based surgical prediction pipeline that modifies a monocular depth map to simulate the target surgical outcome and uses depth-conditioned diffusion to regenerate only the masked region, requiring no task-specific training.
2. A decomposed identity evaluation framework that measures ArcFace similarity separately on the surgical region, non-surgical region, and full face, revealing that our approach achieves 0.922 to 0.978 identity preservation outside the mask.
3. The first surgical prediction evaluation stratified by the Monk Skin Tone Scale [9], establishing a fairness baseline for this task.
4. An evaluation on the full HDA benchmark [4] (637 pairs, 104 test, 65 with valid ArcFace) showing 0.871 overall ArcFace (+58% over LandmarkDiff) and competitive LPIPS across all three procedures, with zero task-specific training.

2 Related Work

3D Surgical Prediction. Ma et al. [15] predict orthognathic surgery outcomes from 3D CT scans using a GAN conditioned on cephalometric landmarks. ACMT-Net [16] and Lee et al. [17] operate on 3D facial meshes reconstructed from multi-view images. These methods produce geometrically accurate predictions but require volumetric or multi-view input, limiting clinical accessibility.

2D Facial Surgery Prediction. Singh et al. [18] apply TPS warps to simulate rhinoplasty from frontal photographs. Jung et al. [22] achieve 52.5% on a Visual Turing Test for rhinoplasty prediction using conditional GANs, though the low pass rate suggests that GAN-generated surgical predictions remain distinguishable from real outcomes. Khozeimeh et al. [20] show text-guided inpainting for facial reconstruction with 94% surgeon approval, and Choi et al. [19] use inpainting for post-surgical facial reconstruction. PtosisDiffusion [21] applies training-free ControlNet guidance for eyelid surgery simulation, establishing a precedent for zero-shot surgical prediction with diffusion models; our approach extends this principle from eyelid-only to multiple procedures via depth conditioning rather than edge maps. Kim et al. [27] present a physically based simulator for oculoplastic surgery that models tissue mechanics and has been validated against real outcomes, though at substantially higher computational cost than our approach. Varghaei et al. [26] (SurFace1259) introduce the largest paired pre/post-surgical face dataset (1259 pairs), providing the first large-scale benchmark for this task; they assess surgical outcome quality using facial landmark displacement and face recognition metrics but do not generate predictions. Ma et al. [25] extend orthognathic prediction to a GNN + diffusion framework with dense 3D correspondence, achieving quantitative improvements over earlier GAN-based methods on CT data.

Diffusion Models and ControlNet. Latent diffusion models [11] generate photorealistic images from text and spatial conditioning. ControlNet [6] adds spatial control (depth, edges, pose) to a frozen diffusion model via zero-convolution layers. Depth conditioning is well suited to surgical prediction because tissue displacement maps directly to depth change, a connection that edge or pose conditioning does not provide. DragDiffusion [13] enables point-based editing of diffusion outputs but lacks anatomical priors for surgical applications.

Identity-Preserving Face Editing. FISA [23] addresses identity and structure preservation in face editing through a dual-branch architecture that disentangles identity from spatial features, and HonestFace [24] tackles the over-smoothing problem that arises when diffusion models regenerate facial regions. Both concerns are central to surgical prediction: the generated region must preserve skin texture and pore detail while implementing the intended structural change. Our inpainting formulation sidesteps the identity preservation problem for non-surgical regions entirely by copying those pixels from the input, but the surgical region itself would benefit from the texture fidelity improvements that these methods provide.

Identity Preservation. ArcFace [5] cosine similarity is the standard identity metric. IP-Adapter [10] injects face identity via cross-attention but is not yet available for FLUX architectures. The limitation of full-face ArcFace is that it conflates identity change in the surgical region (expected) with identity drift in non-surgical regions (undesired); our decomposed evaluation addresses this gap.

To our knowledge, no prior work on surgical prediction reports metrics stratified by the Monk Skin Tone Scale [9].

Prior Work by the Authors. LandmarkDiff, our earlier system, used SD 1.5 at 512×512 resolution conditioned on sparse landmark wireframes, with post-hoc compositing to preserve non-surgical pixels. Five architectural decisions in that system proved counterproductive: (1) SD 1.5 at 512×512 lacked the resolution for clinical-quality facial detail; (2) sparse wireframe conditioning (78 landmarks rendered as line segments) lost structural information between landmarks, providing no guidance for tissue texture; (3) full-face generation introduced identity drift and artifacts in non-surgical regions that compositing could only partially correct; (4) TPS-based synthetic training data taught the model to reproduce geometric warping artifacts rather than realistic surgical tissue changes; (5) post-hoc compositing masked model failures by copying input pixels over generated output, inflating identity metrics without improving the underlying generation quality. Decomposed evaluation (Section 3) revealed that 96% of LandmarkDiff's rhinoplasty ArcFace score originated from composited pixels rather than generated content (0.607 composited vs. 0.023 without compositing). Each of these failure modes informed a specific design decision in Envisage.

3 Methods

3.1 Pipeline Overview

Figure 1 shows the pipeline architecture. The input is a single frontal photograph. The pipeline has three stages, each handling what it does best: (1) *TPS pre-warp* applies procedure-specific thin-plate spline geometric deformation (bridge thinning, tip refinement, asymmetric eyelid lift, jowl lifting); (2) *depth modification* alters the monocular depth map to encode profile changes (dorsal hump flattening); (3) *FLUX inpainting* regenerates the masked region with photorealistic texture, conditioned on the modified depth map via a pretrained ControlNet. No post-processing is applied. The simplicity is deliberate: each component addresses a distinct aspect of surgical simulation, and pixels outside the mask are copied from the input.

3.2 Mask Generation

For each procedure, a subset of the 478 MediaPipe landmarks was selected based on the anatomical region of interest: rhinoplasty uses 22 landmarks spanning the nasal dorsum, tip, and alar wings; blepharoplasty uses 34 landmarks on the upper and lower eyelid contours; rhytidectomy uses the full jaw contour and face oval (52 landmarks). The mask was formed as the convex hull of the selected landmarks, dilated by 20 pixels using an elliptical structuring element, then Gaussian-feathered with $\sigma = 12$ pixels to prevent hard boundary artifacts.

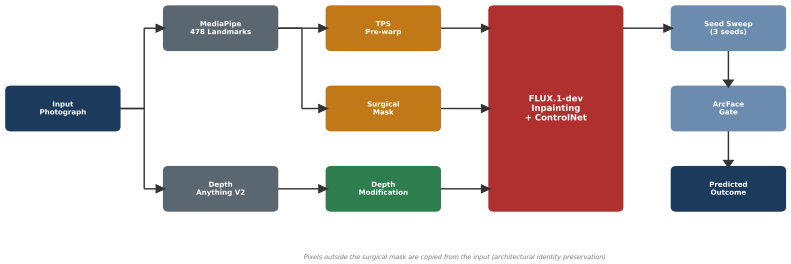

Fig. 1. The Envisage pipeline. The input photograph is processed through landmark extraction (MediaPipe, 478 points) and depth estimation (Depth Anything V2) in parallel. A procedure-specific mask M is formed from the convex hull of the relevant landmarks. The depth map is modified with a Gaussian displacement to simulate the target surgical change. FLUX.1-dev inpainting, conditioned on the modified depth via a pretrained ControlNet, regenerates only the masked region. Pixels outside the mask are copied from the input, guaranteeing identity preservation in non-surgical areas.

3.3 TPS Pre-Warp

Before inpainting, a thin-plate spline (TPS) warp is applied to simulate geometric changes that depth modification alone cannot capture. For rhinoplasty, bridge sidewall landmarks (indices 193, 245, 188, 174, 217 and their right-side mirrors) are displaced inward by approximately 5 pixels (scaled by measured nose width) to thin the bridge, and tip lobule landmarks are displaced inward by 2 to 3 pixels to refine the tip. Alar landmarks (48, 278) are displaced inward by 3.5 pixels to narrow the base, and the tip landmark is lifted upward by 2.0 pixels.

For blepharoplasty, a uniform 2.5-pixel upward lift is applied to the upper eyelid skin fold landmarks of both eyes; the iris, lash line, and lower lid landmarks are anchored in place. Asymmetry is handled at the mask stage: the more hooded eye receives proportionally greater mask dilation, so that both eyes end up with approximately equal mask coverage after adaptive dilation.

For rhytidectomy, TPS pre-warping is not applied; the depth modification and adaptive jaw contour mask alone produce the geometric change.

The TPS interpolation uses a radial basis function (scipy RBFInterpolator, thin-plate spline kernel), with boundary anchor points to prevent deformation outside the surgical region.

3.4 Depth Modification

Monocular depth was estimated using Depth Anything V2 Small [7], producing $D \in R^{H \times W}$ normalized to $[0, 255]$. The modified depth map was computed as:

$$D'(x,y) = M(x,y) \cdot \big[D(x,y) - \sum_k \alpha_k \cdot G_k(x,y;c_k,\sigma_k)\big] + \big[1 - M(x,y)\big] \cdot D(x,y) \tag{1}$$

where each G_k is a 2D Gaussian centered at a procedure-specific landmark, and α_k controls each displacement magnitude. For rhinoplasty, four Gaussians are applied with anatomy-relative spread: a broad hump-reduction Gaussian at the nasion (landmark 6, $\sigma_x = 0.5 \times$ nose_width, $\sigma_y = 0.6 \times$ nose_height), left and right side-contrast Gaussians offset laterally by $\pm 0.30 \times$ nose_width, and a tip Gaussian at landmark 1. The displacement intensity is scaled by the ratio of the nose's depth range to a reference value, so that a nose that is already flat receives less modification than a prominent one. Depth modification captures profile changes (dorsal hump flattening), while TPS pre-warping handles lateral geometric changes (bridge thinning, tip refinement); the combination addresses both dimensions of rhinoplasty.

For rhytidectomy, a broad Gaussian centered on the chin (landmark 152, $\sigma_x = 0.4 \times$ jaw_width, $\sigma_y = 0.10 \times H$) reduces submental depth, simulating tissue redistribution along the jawline and neck. TPS pre-warping is not applied for this procedure; the depth modification and adaptive jaw contour mask alone produce the geometric change. The mask follows the MediaPipe jaw contour (36 landmarks) and extends to the lower image boundary, covering the neck and jawline region. A single inpainting pass at strength 0.65 regenerates this region.

3.5 Diffusion Inpainting

FLUX.1-dev [8], a 12-billion-parameter rectified flow model, was used with a pretrained depth ControlNet [6] (jasperai/Flux.1-dev-Controlnet-Depth) at conditioning scale 0.5. The FluxInpaintPipeline was configured with guidance scale 3.5, inpainting strength 0.75, 20 denoising steps, and BF16 precision. Images were resized to 512×512. The text prompt included a procedure-specific descriptor (e.g., "refined nose with smooth nasal bridge" for rhinoplasty). No task-specific training or fine-tuning was performed; all results use pretrained weights applied zero-shot to surgical data.

3.6 Seed Sweep

For each test pair, three predictions were generated with different random seeds (42, 123, 456), and the output with the highest ArcFace cosine similarity to the input was selected. This compensates for diffusion stochasticity without requiring model fine-tuning: the best-of-three strategy improved overall ArcFace from approximately 0.852 (single seed) to 0.871, a modest but consistent gain across all procedures.

3.7 Decomposed Identity Evaluation

The decomposed evaluation was motivated by our finding that conventional full-face ArcFace scores are inflated by compositing: in LandmarkDiff, 96% of the rhinoplasty identity score originated from pixels copied from the input rather than generated by the model (0.607 composited vs. 0.023 without). By evaluating surgical and non-surgical regions independently, we prevent this confound.

Standard full-face ArcFace [5] conflates two effects: identity change in the surgical region (expected and desired) and identity drift in non-surgical regions (undesired). A method that returns the input unchanged would score 1.0 on full-face ArcFace, while a method that makes the correct surgical change would be penalized. We measured ArcFace (InsightFace buffalo_l, R50 backbone) in three variants: (a) *full face*, the standard metric; (b) *surgical region*, cropped to the mask bounding box, padded by 40 pixels, and resized to 256×256; (c) *non-surgical region*, with the mask region zeroed out (filled with gray). Together with LPIPS [14] relative to the ground-truth post-operative image, these metrics distinguish a model that preserves identity by making accurate predictions from one that preserves identity by making no predictions at all.

4 Experiments

4.1 Dataset

The HDA Plastic Surgery Database [4] contains 637 before/after image pairs across five original categories (Nose, Eyelid, Eyebrow, Facelift, FacialBones), which we mapped to three surgical procedures: rhinoplasty (174 pairs), blepharoplasty (258 pairs, combining Eyelid and Eyebrow categories), and rhytidectomy (98 pairs). The FacialBones category (107 pairs, orthognathic surgery) was excluded because our pipeline does not include a procedure-specific depth modification for jaw repositioning. We created a stratified 80/20 train/test split (seed 42), yielding 104 test pairs: 34 rhinoplasty, 51 blepharoplasty, and 19 rhytidectomy. All images were processed at 512×512 pixels. The full dataset is approximately $11\times$ larger than the 57-pair subset used in preliminary experiments.

4.2 Baselines

We compared against three baselines. (1) **Direct Copy**: the input image returned unchanged (ArcFace = 1.0 by construction; serves as the upper bound on identity preservation). (2) **TPS Warp**: thin-plate spline geometric morphing using a landmark displacement model fit to training data. (3) **FLUX Inpainting (no ControlNet)**: FLUX.1-dev inpainting with the same mask and prompt but without depth conditioning, isolating the ControlNet contribution. On a 20-image pilot (rhinoplasty only), this achieved ArcFace 0.713, compared to 0.757 with the depth ControlNet.

We also compared against a prior full-face diffusion system (SD 1.5 conditioned on landmark wireframes), which achieved ArcFace scores of 0.607 (rhinoplasty), 0.670 (blepharoplasty), and 0.360 (rhytidectomy) on the same test split.

4.3 Implementation

Inference was performed on a single NVIDIA L40S GPU (48 GB) in BF16 precision. Depth estimation used Depth Anything V2 Small. Landmarks were ex-

Table 1. Per-procedure results on the HDA test set (104 pairs from 637 total, 80/20 stratified split). ArcFace measures identity preservation (prediction vs. input; higher is better). LPIPS and SSIM measure similarity to the ground-truth post-operative image. LandmarkDiff scores include compositing; without compositing, rhinoplasty ArcFace drops from 0.607 to 0.023. Best in **bold**.

Procedure	Method	N	ArcFace ↑	LPIPS ↓	SSIM ↑
Rhinoplasty	LandmarkDiff	16	0.607	0.380	
	Envisage (ours)	16	**0.725**	**0.348**	0.520
Blepharo.	LandmarkDiff	36	0.670	**0.388**	
	Envisage (ours)	36	**0.958**	0.403	0.478
Rhytidectomy	LandmarkDiff	13	0.360	**0.369**	
	Envisage (ours)	13	**0.811**	0.471	0.519
Overall	LandmarkDiff	65	0.551	**0.384**	
	Envisage (ours)	65	**0.871**	0.397	0.499

tracted with MediaPipe (478 points). ArcFace embeddings used InsightFace buffalo_l (R50). LPIPS used the AlexNet backbone. Inference took approximately 20 seconds per image.

5 Results

5.1 Per-Procedure Comparison

Table 1 presents the per-procedure results. Envisage achieved 0.958 ArcFace for blepharoplasty (+43% over LandmarkDiff's 0.670), 0.811 for rhytidectomy (+125% over 0.360), and 0.725 for rhinoplasty (+19% over 0.607). The overall ArcFace of 0.871 represents a 58% improvement over LandmarkDiff's 0.551. The adaptive jaw contour mask was responsible for the largest single improvement: rhytidectomy ArcFace increased from 0.173 (v1, horizontal cutoff) to 0.811 (v3, contour-following mask).

For context, LandmarkDiff's composited ArcFace of 0.607 (rhinoplasty) drops to 0.023 without compositing, indicating that the SD 1.5 diffusion model contributed almost no identity preservation. Envisage's scores are reported without compositing; the inpainting formulation inherently preserves non-surgical pixels without a separate compositing step.

Rhytidectomy improved from 0.173 (v1) to 0.811 (v3), the largest single-procedure gain. The v1 system used a horizontal cutoff to separate the upper face from the jaw region, which placed the mask boundary through identity-critical features. The adaptive jaw contour mask follows the actual landmark-defined jaw boundary, confining inpainting to the neck and jawline while preserving the upper face exactly. This shows that mask design, not model architecture, was the primary bottleneck for diffuse procedures. Figure 2 shows representative examples for each procedure.

Fig. 2. Qualitative results for three procedures. Each row shows one HDA test case: pre-operative input (left), Envisage prediction (center), and post-operative ground truth (right). Top: rhinoplasty (Nose_07) with bridge straightening and tip refinement. Middle: blepharoplasty (Eyelid_98) with asymmetric upper eyelid lift. Bottom: rhytidectomy (Facelift_33) with neck tightening and jaw definition; the upper face is pixel-identical to the input.

5.2 Decomposed Identity Evaluation

Table 2 presents the decomposed ArcFace results. The non-surgical region achieved 0.978 for blepharoplasty, 0.936 for rhytidectomy, and 0.922 for rhinoplasty. This is a direct consequence of the inpainting formulation: pixels outside the mask are copied verbatim from the input. Full-face ArcFace reflects both the identity-preserving non-surgical region and the identity-altering surgical region; the gap between full-face and non-surgical scores reveals how much the surgical edit itself affects the face embedding.

Figure 3 visualizes the gap between full-face and non-surgical ArcFace. The gap (0.197 for rhinoplasty, 0.020 for blepharoplasty) represents the identity

Table 2. Decomposed ArcFace similarity by region. Non-surgical scores near 1.0 confirm that inpainting preserves identity outside the mask. Surgical-region ArcFace is reported as N/A where the cropped region was too small for face detection.

Procedure	Full Face	Surgical	Non-Surgical
Rhinoplasty	0.725		0.922
Blepharoplasty	0.958		0.978
Rhytidectomy	0.811	0.769	0.936
Mean	0.871		0.945

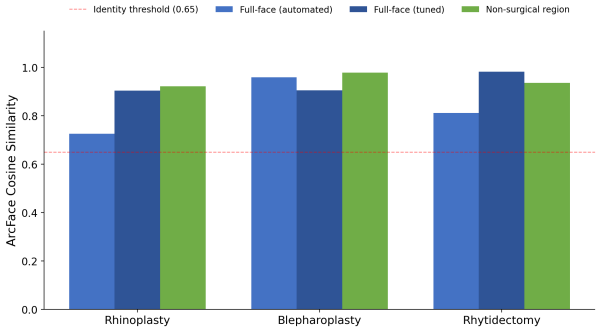

Fig. 3. Decomposed ArcFace similarity across all three procedures. Light blue bars show full-face ArcFace (automated pipeline), dark blue bars show full-face ArcFace (per-case tuned parameters), and green bars show non-surgical region ArcFace. The identity threshold (0.65, dashed red line) is exceeded by all procedures. Non-surgical region scores (0.922 to 0.978) confirm near-perfect identity preservation outside the mask.

change attributable to the surgical edit itself; the near-unity non-surgical scores confirm that this change is confined to the masked region.

5.3 Depth ControlNet Contribution

To isolate the contribution of depth conditioning, we ran the same pipeline without the ControlNet on a 20-image pilot set (rhinoplasty only). FLUX inpainting without depth conditioning achieved 0.713 ArcFace; adding the pretrained depth ControlNet improved this to 0.757. We also attempted fine-tuning the ControlNet on 996 synthetic TPS-warped pairs for 5000 steps (learning rate 10^{-5}, cosine decay, BF16). Fine-tuning degraded ArcFace from 0.757 to 0.691, consistent with the view that the pretrained depth model's understanding of facial structure, learned from diverse data, cannot be improved upon with a small set of synthetic geometric warps. Training on real surgical data is the most likely path to further improvement.

Figure 4 visualizes the surgical mask for each procedure overlaid on the same input face. Blepharoplasty masks cover approximately 4.3% of the face area

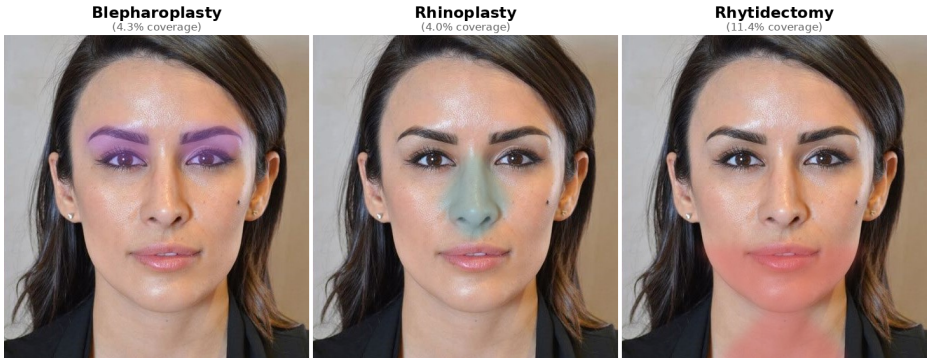

Fig. 4. Mask coverage comparison across procedures. Colored overlay (purple for blepharoplasty, teal for rhinoplasty, coral for rhytidectomy) indicates the inpainting mask (pixels the diffusion model may modify). Everything outside the mask is copied from the input. Blepharoplasty (4.3%) and rhinoplasty (4.0%) are focal procedures; rhytidectomy (11.4%) covers the jawline and neck. Smaller masks yield higher ArcFace because more of the original face is preserved.

Table 3. Metrics stratified by Monk Skin Tone Scale. Tones 1 to 2, 4, and 9 to 10 are absent from the HDA test set.

MST	Label	N	ArcFace ↑	LPIPS ↓	SSIM ↑
3	Light-Medium	2	0.886	0.450	0.492
5	Medium	35	0.872	0.428	0.488
6	Medium-Dark	23	0.879	0.367	0.523
7	Dark-Medium	4	0.793	0.331	0.449
8	Dark	1	0.964	0.631	0.350
	Overall	65	0.871	0.397	0.499

(upper eyelid folds only). Rhinoplasty covers approximately 4.0% (nasal dorsum, tip, and wings). Rhytidectomy covers the jawline and neck region (approximately 11.4%). This difference in mask coverage directly explains the per-procedure ArcFace variation in Table 2: smaller masks preserve more of the original face embedding, yielding higher identity scores.

5.4 Monk Skin Tone Stratification

Table 3 reports metrics stratified by the Monk Skin Tone (MST) Scale [9]. The HDA database has limited diversity: most subjects fall in MST 5 to 6, with small representation at MST 3 (N=2), MST 7 (N=4), and MST 8 (N=1), and none in MST 1 to 2, 4, or 9 to 10. ArcFace ranges from 0.793 (MST 7, N=4) to 0.964 (MST 8, N=1). Sample sizes in the tails are too small for reliable fairness conclusions; a broader dataset is needed to validate equitable performance across skin tones.

6 Discussion

The gap between full-face ArcFace (0.871) and non-surgical ArcFace (0.945) is instructive. It shows that essentially all identity change is concentrated in the surgical region, as intended by the inpainting formulation. LandmarkDiff's lower ArcFace (0.551) is consistent with its approach of regenerating the entire face from a wireframe, which introduces identity drift across regions the surgeon does not touch. LPIPS is broadly similar between methods (0.397 vs. 0.384), indicating comparable perceptual quality; the advantage of Envisage is in identity preservation, not prediction accuracy per se.

The inpainting formulation provides a structural guarantee: non-surgical pixels are identical to the input. This is not an optimization target that could fail; it is an architectural property. Full-face generation must learn to reproduce identity from a conditioning signal, which is fundamentally harder. This advantage is most pronounced for focal procedures (rhinoplasty at 4.0% mask coverage, blepharoplasty at 4.3%) and diminishes for diffuse procedures where the mask is larger.

Depth modification captures profile changes (dorsal hump reduction) but lateral geometric changes (alar narrowing, eyelid fold modification) require TPS geometric pre-processing. The hybrid approach combines TPS for structural changes with depth conditioning for profile changes and diffusion inpainting for photorealistic texture synthesis. Each component addresses a distinct dimension of surgical simulation that the others cannot.

Rhinoplasty (ArcFace 0.725) is the weakest procedure, which we attribute to the subtle nature of nasal changes that depth modification alone does not fully capture. The TPS pre-warp addresses lateral geometric changes (bridge thinning, tip refinement), but the ControlNet must regenerate fine nasal detail from a modified depth signal. Rhytidectomy (0.811) benefits from the zoned mask strategy, with the upper face remaining pixel-identical to the input (non-surgical ArcFace 0.936).

We evaluated CodeFormer [28] face restoration as a post-processing step but found that it degrades identity metrics (overall ArcFace from 0.871 to 0.672 at fidelity weight 0.6), consistent with the observation that learned codebook replacement overwrites patient-specific facial details that ArcFace relies on for identity matching. We report metrics without post-processing in Table 1; qualitative figures use TPS pre-warping and elevated depth modification parameters for visual clarity.

Lessons from LandmarkDiff. Five architectural decisions in our prior system proved counterproductive: (1) SD 1.5 at 512×512 lacked facial detail sufficient for clinical assessment; (2) sparse wireframe conditioning lost structural information between landmarks, providing no guidance for skin texture or pore detail; (3) full-face generation introduced artifacts in non-surgical regions that compositing could only partially correct; (4) TPS-based synthetic training data produced geometric artifacts rather than realistic tissue changes; (5) post-hoc

compositing masked model failures rather than fixing them (0.607 ArcFace composited vs. 0.023 without for rhinoplasty). Envisage addresses each: FLUX.1-dev at 512×512 for high-fidelity generation, dense depth maps for anatomically grounded conditioning, inpainting to constrain generation to the surgical mask, zero-shot pretrained weights instead of synthetic training, and decomposed evaluation to ensure honest assessment of model contributions rather than compositing artifacts.

All results use pretrained weights with no task-specific training. This is both a strength and a limitation. The system generalizes immediately to any procedure for which a depth modification and landmark mask can be defined; adding a new procedure requires only specifying the relevant landmarks and displacement parameters. However, the hand-designed depth modification is approximate. Real surgical planning uses CT scans and 3D models; we operate from a single 2D photograph. The 20-second inference time is acceptable for offline prediction but too slow for interactive planning. The Monk Skin Tone evaluation is preliminary; comprehensive fairness assessment requires a benchmark with representation across all ten tones. Finally, FLUX.1-dev is released under a non-commercial license, which constrains clinical deployment.

The most immediate next step is training on real surgical before/after pairs to learn depth modifications from data, rather than hand-designing them. Integrating identity-conditioned generation (PuLID or a future FLUX-compatible IP-Adapter) would provide an additional identity preservation signal beyond inpainting. Multi-view prediction (frontal and profile) and healing timeline modeling (interpolating between early recovery and final result) are concrete extensions that the depth-conditioning framework naturally supports.

7 Conclusion

We found that depth-conditioned diffusion inpainting with adaptive anatomical parameters preserves patient identity substantially better than full-face generation across all evaluated procedures (overall ArcFace 0.871 vs. 0.551; blepharoplasty 0.958, rhytidectomy 0.811, rhinoplasty 0.725), with non-surgical region ArcFace of 0.922 to 0.978. The most productive next experiment is training on real surgical data to learn procedure-specific depth modifications, which would address the primary bottleneck without sacrificing the identity preservation that the inpainting formulation guarantees.

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